Teacher Wu Kuan | Exam Trap #1 — Radical Equations: Check, Don't Divide

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1. The trap question

Solve $\sqrt{3x + 1} - \sqrt{x - 1} = 2$. Find the sum of all real solutions.

2. The method

• Domain first: $x \geq 1$.
• Square twice to clear the roots → candidates $x = 1$ and $x = 5$.
• Check each in the ORIGINAL equation: $x = 1 \to 2 - 0 = 2$ ✓; $x = 5 \to 4 - 2 = 2$ ✓. Sum = 6.
• Trap: dividing by $\sqrt{x - 1}$ (which can be 0) loses $x = 1$. Squaring can ADD fake roots, so always check.

3. Parent check method

Ask your child to show the check step for every candidate answer — no check, no credit.

Practice

1. Solve $\sqrt{x + 3} = x - 3$.
2. Solve $\sqrt{2x + 1} = x - 1$.
3. Solve $x^2 = 3x$. (Don't divide by x!)
4. Solve $\sqrt{x + 5} = x + 3$.
5. Find the sum of all real solutions of $\sqrt{3x + 1} - \sqrt{x - 1} = 2$.

Answers

1. $x = 6$ ($x = 1$ is extraneous: $\sqrt{4} = 2 \neq -2$)
2. $x = 4$ ($x = 0$ is extraneous)
3. $x = 0$ or $x = 3$
4. $x = -1$ ($x = -4$ is extraneous)
5. 6 ($x = 1$ and $x = 5$ both check)

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